MODULE 4 | AIRCRAFT FLIGHT CONTROLS

ASYS 245 • Part B • Cornell Notes • SECTION 2

Flaps • slots and slats • spoilers • speed brakes

CUES / QUESTIONS

1. How do flaps and slats affect lift and drag?

2. What are the four common flap types?

3. What is the difference between a slot and a slat?

4. What happens when spoilers deploy?

5. How do speed brakes slow the aircraft?

NOTES | HIGH-LIFT AND DRAG DEVICES

1 FLAPS + SLATS: CHANGING AIRFLOW



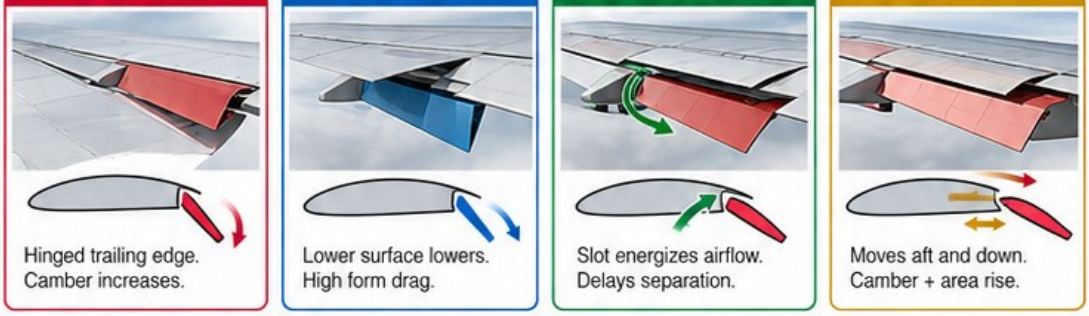
Flaps increase wing camber and, for Fowler designs, wing area. They increase lift and drag. Slats and slots guide higher-energy air along the upper surface, delaying separation at high angle of attack.

LIFT INCREASES

DRAG INCREASES

Landing: steeper approach without excess acceleration and lower-speed control.

2 FOUR COMMON FLAP TYPES



PLAIN FLAP	SPLIT FLAP	SLOTTED FLAP	FOWLER FLAP
Hinged trailing edge. Camber increases.	Lower surface lowers. High form drag.	Slot energizes airflow. Delays separation.	Moves aft and down. Camber + area rise.

3 SLOTS vs. SLATS



Slot: an opening that allows air to pass from below to the upper wing surface. **Slat:** a movable leading-edge surface that forms such a duct. The slides describe some automatic slats extending at high angle of attack.

4 SPOILERS



Spoilers reduce lift and increase drag. One-sided use can assist roll. Symmetric flight-spoiler use supports descent; ground spoilers dump wing lift after touchdown.

5 SPEED BRAKES AND KEY DISTINCTIONS

Speed brakes are deployable surfaces that increase drag to slow an aircraft in flight or during landing. They are positioned and operated to avoid unwanted pitch, roll or yaw moments.

SPEED BRAKES (FLIGHT / DESCENT)



- Increase drag
- Decrease lift
- Used in flight (descent)
- Usually on upper and lower wing surfaces

GROUND SPOILERS (AFTER TOUCHDOWN)



- Dump wing lift
- Increase drag
- Deploy automatically or with lift-sensor
- Assist braking on rollout

FLAPS: increase lift + drag | **SLOTS/SLATS: energize airflow, delay separation** | **SPOILERS: decrease lift** | **SPEED BRAKES: add drag**

SECTION 2 | STUDENT SUMMARY

- Flaps increase camber (and sometimes area), generating more lift and drag.
- Slats and slots energize airflow to help delay separation near stall.
- Spoilers decrease wing lift and increase drag, including after touchdown.
- Speed brakes mainly add drag and are balanced to avoid unwanted moments.